

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		SLOT B
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	CSE AI	Date:	19-08-2026

Code of Conduct

I P Mohan Babu / 192472254 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P Mohan Babu

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. Source Sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.

2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Assessment Tool 1 – Constraint-Based Problem Solving

Q1. Discourse, Reference Resolution & Text Coherence

Given paragraph

“John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home.”

(a) Referring expressions and possible antecedents

Mention	Type	Possible antecedents
He	Pronoun	John, Mary
She	Pronoun	John, Mary
it	Pronoun	ball, park
him	Pronoun	John, Mary, dog
they	Pronoun	John, Mary, dog
all	Group reference	John, Mary, dog

(b) Applying constraints

1. Gender and number agreement

- **He** → **John** because John is male and singular.
- **She** → **Mary** because Mary is female and singular.
- **it** → **ball** because *ball* is singular and inanimate.
- **him** → **John** is grammatically possible because John is male, but semantic and discourse constraints must also be considered.
- **they** → **John + Mary + dog** because the plural pronoun can refer to the complete group.

2. Recency

Recency prefers the closest suitable antecedent. However, it should not override stronger agreement or semantic constraints.

For example, after “The dog chased him,” the immediately preceding candidate is **dog**, but **him** cannot normally refer to dog because *him* requires a masculine human/male referent in this context. Therefore, **John** remains the best antecedent.

3. Semantic compatibility

- A person can **bring a ball** → John is compatible with “He brought a ball.”
- A person can **want to play with a ball** → Mary is compatible with “She wanted to play with it.”
- A dog can **chase a person** → “him” can naturally refer to John.
- John, Mary and the dog can **go home together** → “they all” refers to the complete group.

4. Coherence

The resulting entity chains must remain consistent:

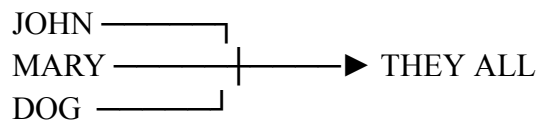
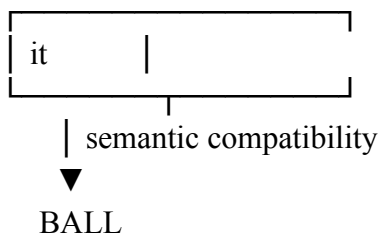
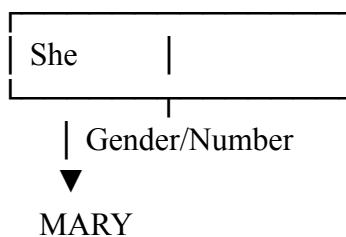
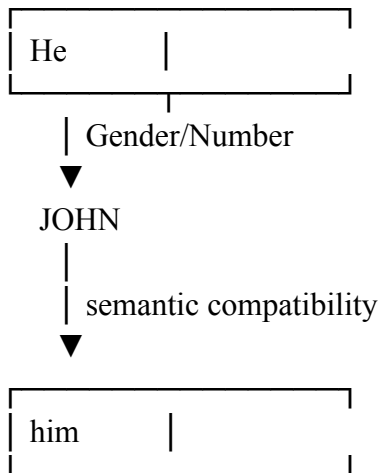
- **John** → **He** → **him** → **they**
- **Mary** → **She** → **they**
- **ball** → **it**
- **dog** → **they**

(c) Constraint table

Variable	Initial domain	Constraints applied	Final value
He	John, Mary	Gender + number + semantics	John
She	John, Mary	Gender + number	Mary
it	ball, park	Number + semantic compatibility	ball

Variable	Initial domain	Constraints applied	Final value
him	John, Mary, dog	Gender + animacy + action suitability + coherence	John
they	John + Mary; John + Mary + dog	Number + discourse coherence + “all”	John + Mary + dog

Constraint graph



(d) Final coreference chains

- **John** → **He** → **him** → **they**
- **Mary** → **She** → **they**
- **ball** → **it**
- **dog** → **they**

A clear rewritten version is:

“John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The

dog chased John excitedly. Finally, John, Mary, and the dog all went home.”

(e) Priority order and effect of relaxing recency

A suitable priority order is:

Agreement → Semantic compatibility → Coherence → Recency

Agreement and semantic compatibility should receive higher priority because they eliminate grammatically or conceptually impossible references. Coherence ensures that the same entities are tracked consistently throughout the discourse. Recency is useful for selecting among otherwise valid candidates, but it should not override stronger constraints.

If recency is relaxed, resolution can still succeed because gender, semantic compatibility and discourse coherence provide sufficient evidence. However, when several candidates satisfy the stronger constraints, the system may have more ambiguity and require additional contextual information.

Q2. Dialog and Conversational Agents

Given conversation

User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required dialog acts: Advise + Encourage

(a) Discourse planning steps

The following planning steps are appropriate:

1. **Identify the user’s goal/state:** The user has an exam tomorrow and has difficulty concentrating.
2. **Select dialog acts:** Provide practical advice and encouragement.
3. **Maintain entity coherence:** Continue referring to the *exam*, *concentration*, and *you*.
4. **Establish discourse relation:** Use a Cause–Effect or Elaboration relation.
5. **Select content:** Suggest focused study sessions and short breaks.
6. **Apply constraints:** Include at least two keywords: *focus*, *break*, and/or *confident*.
7. **Surface realization:** Produce a polite, positive response of exactly 2–3 sentences.

(b) Three valid responses

Response 1:

“Since your exam is tomorrow, try studying in short focused sessions and take a short break whenever your concentration drops. This can help you regain focus and feel more confident about the exam.”

Response 2:

“Your exam is important, so start with one small topic and give yourself a short break after each focused study session. You can improve your concentration step by step, and you should feel confident about what you have prepared.”

Response 3:

“Because your exam is tomorrow, put away distractions and focus on one topic at a time, taking a short break when needed. Remember that steady preparation can help you concentrate better and become more confident.”

(c) Evaluation and best response

Criterion	Response 1	Response 2	Response 3
Entity coherence	✓	✓	✓
Advice + encouragement	✓	✓	✓
Cause–Effect relation	✓	✓	✓
At least 2 keywords	✓	✓	✓
2–3 sentences	✓	✓	✓
Logical consistency	✓	✓	✓
Positive/polite tone	✓	✓	✓

Best response: Response 1.

It is the strongest because the relationship between the exam, concentration problem, study method and expected improvement is very clear. It provides concrete advice while also encouraging the student. It satisfies all the constraints without sounding repetitive.

(d) Effect of violating two constraints**Violation 1 – Response length constraint:**

If the response contained five or six sentences instead of 2–3, it would become less concise and could overwhelm a student who is already struggling to concentrate.

Violation 2 – Entity coherence constraint:

If the response suddenly discussed unrelated topics instead of the exam and concentration, the conversational context would become inconsistent. The advice would appear less relevant and less useful.

Q3. Machine Translation, WSD, Predicate Logic & Discourse**Source sentence**

“The bank by the river flooded after the storm, but it was saved by quick action.”

(a) Word Sense Disambiguation

The correct sense of **bank** is **riverbank**, not financial institution.

Constraint reasoning

1. **Context constraint:** The phrase “by the river” strongly indicates a geographical riverbank.
2. **Semantic constraint:** A riverbank can flood; a financial institution normally does not flood in the intended literal sense.
3. **Relation constraint:** “flooded” is naturally associated with a riverbank and water.
4. **Discourse coherence:** The pronoun **it** refers back to the riverbank.

Therefore:

bank = riverbank

The intended interpretation is:

“The riverbank near the river flooded after the storm, but the riverbank was saved from further damage by quick action.”

(b) Predicate logic representation

Let:

- $River(r) = r$ is a river
- $Bank(b) = b$ is a bank/riverbank
- $Storm(s) = s$ is a storm
- $location(b,r) =$ bank b is located beside river r
- $Flood(b,s) =$ bank b flooded after storm s
- $QuickAction(q) = q$ is a quick action
- $Saved(b,q) = b$ was saved/protected by q

A simplified representation is:

$\exists b \exists r \exists s \exists q [$

$Bank(b) \wedge$

$River(r) \wedge$

$Storm(s) \wedge$

$location(b,r) \wedge$

$Flood(b,s) \wedge$

QuickAction(q) $\wedge$

Saved(b,q)

]

The temporal relation can additionally be represented as:

after(Flood(b), Storm(s))

The discourse relation between the two clauses is:

Contrast(

Flood(b, s),

Saved(b, q)

)

Thus, the complete semantic interpretation captures the entity, location, flooding event, storm, saving action and contrast relation.

(c) Target-language sentence / simple English paraphrase

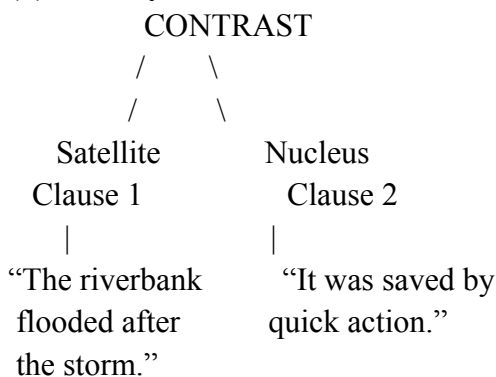
Simple English paraphrase:

“The riverbank beside the river flooded after the storm, but quick action prevented it from suffering further damage.”

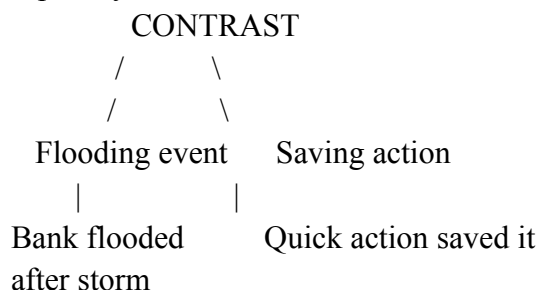
This preserves:

- the riverbank entity,
- the river location,
- the storm,
- the flooding event,
- the quick action,
- the result of the action,
- and the contrast relation introduced by **but**.

(d) RST-style discourse tree



More explicitly:



The word **“but”** signals the contrast between the negative event (flooding) and the positive outcome (being saved through quick action).

(e) Advantages of a constraint-based approach over pure statistical MT

A constraint-based approach has several advantages for this sentence:

1. **Better word-sense disambiguation:**
Contextual constraints identify *bank* as a riverbank rather than a financial institution.
2. **Semantic consistency:**
The selected sense must be compatible with the predicate **flood** and the phrase **by the river**.
3. **Coreference preservation:**
The system can ensure that **it** refers to the same riverbank entity.
4. **Discourse preservation:**
The contrast relation expressed by **but** can be explicitly maintained in the output.
5. **Information preservation:**
Important entities and relations such as river, storm, flooding and quick action are less likely to be lost.
6. **Interpretability:**
The decision can be explained using explicit constraints instead of relying only on statistical probabilities.
7. **Controlled output:**
Predicate logic and semantic constraints help prevent translations or paraphrases that are grammatically fluent but semantically incorrect.

A pure statistical or neural translation system may produce a fluent sentence, but without explicit constraints it could potentially select an incorrect sense of *bank*, weaken the coreference of *it*, or alter the contrast relation. Constraint-based processing therefore provides stronger control over semantic correctness and discourse coherence.

Overall Constraint Priority

For all three questions, a useful general strategy is:

Hard constraints



Agreement / grammatical validity



Semantic compatibility



Coreference and entity consistency



Discourse coherence



Recency / preference constraints



Surface fluency

Hard constraints should eliminate impossible interpretations first. Softer constraints such as recency should then rank the remaining valid alternatives. This produces solutions that are both linguistically valid and contextually coherent.